POCKETSMART AI

Your Smart Budget & Recommendation Assistant

A Project Report

Submitted in partial fulfillment of the requirements for the academic project

Academic Year 2026

Prepared by: Team Leader S Thilagavathi

Department: Computer Application

College: TJS C COLLEGE OF ARTS AND SCIENCE

CERTIFICATE

This is to certify that the project report entitled “PocketSmart AI – Your Smart Budget & Recommendation Assistant” is a bonafide work carried out Team Leader S Thilagavathi under the guidance and supervision of the undersigned during the academic year 2026.

The project demonstrates the design and development of an intelligent personal budgeting assistant that helps users record expenses, understand spending patterns, create budgets, and receive personalized recommendations.

Team Members	**Name / Designation**
S Thilagavathi	Project Guide S Thilagavathi
S Sanjana	Head of Department Sivaranjani R
N Tamilselvi V Thenmozli	External Examiner

DECLARATION & ACKNOWLEDGEMENT

DECLARATION

I hereby declare that the project entitled “PocketSmart AI – Your Smart Budget & Recommendation Assistant” is my original academic work. The project has been prepared for educational purposes and has not been submitted elsewhere for the award of any other qualification.

ACKNOWLEDGEMENT

I express my sincere gratitude to my project guide, faculty members, department, institution, friends, and family for their support and encouragement throughout the development of this project.

I also thank the open-source software community and documentation resources that helped in understanding web development, databases, data analysis, and artificial intelligence concepts used in this project.

ABSTRACT

PocketSmart AI is a smart personal budgeting and recommendation assistant designed to help users manage everyday finances in a simple and understandable way. Many people record expenses irregularly and find it difficult to identify unnecessary spending, maintain a monthly budget, or decide how much they can safely spend.

The proposed system provides a centralized platform where users can add income and expenses, categorize transactions, define monthly budgets, view spending summaries, and receive recommendations. An AI-based recommendation layer analyzes transaction patterns and generates practical suggestions such as reducing overspending in selected categories, setting category limits, and planning future expenses.

The system can use a web-based front end, a server-side application, a database, and a recommendation engine. Data visualization helps users understand their financial behavior through charts and summaries. The system is designed with privacy, simplicity, and responsible recommendations in mind.

Keywords: Personal Finance, Budgeting, Artificial Intelligence, Recommendation System, Expense Tracking, Data Visualization, Machine Learning.

TABLE OF CONTENTS & PROJECT OVERVIEW

Chapter / Section	Page
1. Introduction	6
2. Problem Statement & Objectives	7
3. Existing and Proposed System	8
4. Requirements & Feasibility	9
5. System Architecture & Modules	10
6. Database Design	11
7. AI Recommendation Methodology	12
8. User Interface & Workflow	13
9. Testing, Security & Privacy	14
10. Results, Conclusion & Future Scope	15

PROJECT OVERVIEW

PocketSmart AI combines traditional expense management with intelligent recommendations. The user enters or imports transaction information. The system stores the data, calculates financial summaries, detects patterns, and produces recommendations based on configurable rules and optional machine-learning models.

1. INTRODUCTION

Personal finance management is an important everyday activity. A person may have several sources of income and many small expenses such as food, transportation, education, shopping, subscriptions, and entertainment. Without a structured system, these expenses can be difficult to track.

PocketSmart AI is proposed as a digital assistant that makes budgeting easier. Instead of only displaying transaction records, the system attempts to explain spending behavior and provide useful recommendations.

1.1 Need for the System

- Users need a simple way to record and categorize expenses.
- Users need to compare actual spending with planned budgets.
- Users need alerts when spending approaches a category limit.
- Users need summaries that are easy to understand.
- Users can benefit from personalized suggestions based on their own transaction history.

1.2 Scope

The project focuses on personal budgeting. It can support income and expense tracking, category budgets, dashboard reports, recommendation generation, and basic alerts. It is not intended to replace a bank, financial institution, accountant, or licensed financial adviser.

2. PROBLEM STATEMENT & OBJECTIVES

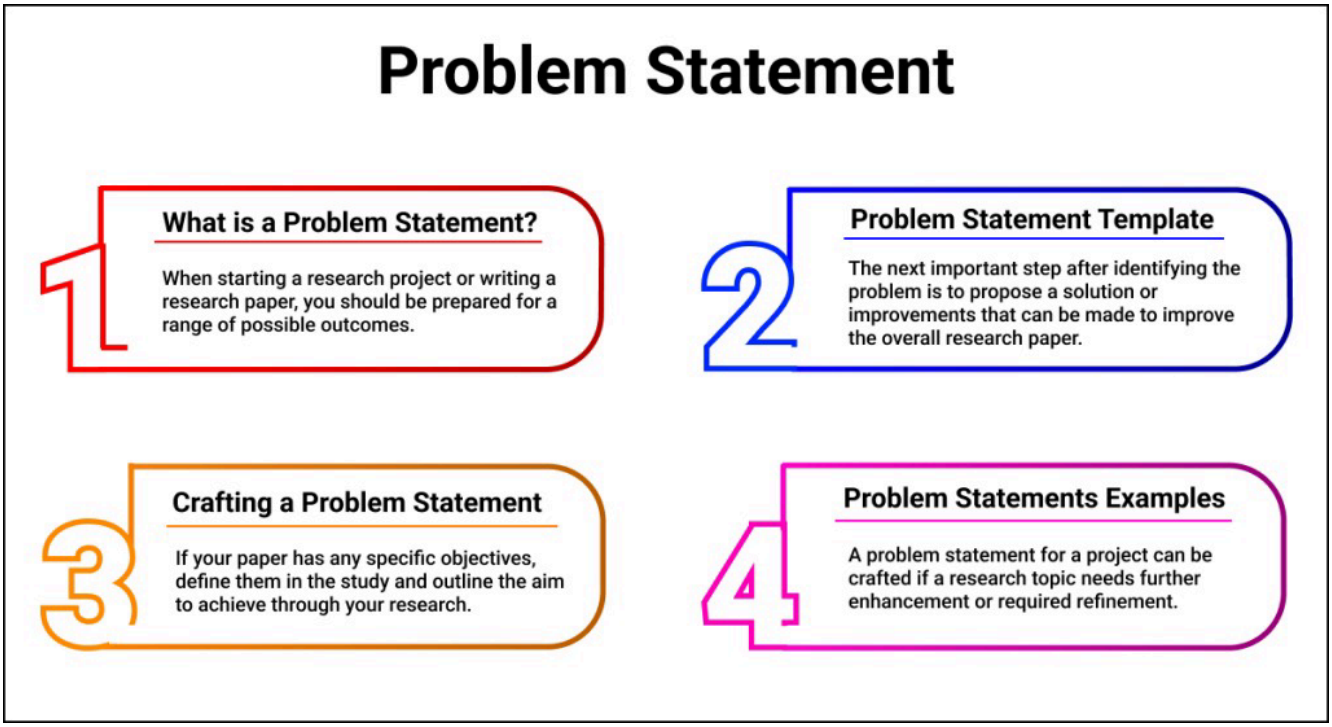
2.1 Problem Statement

Existing expense-tracking approaches often require users to manually review many transactions and calculate totals themselves. A basic expense tracker may show what was spent but may not clearly explain spending trends or suggest practical next steps. PocketSmart AI addresses this gap by combining budgeting, analytics, and recommendation features in one application.

2.2 Objectives

- Develop a user-friendly personal budget management system.
- Allow users to record income and expenses.
- Categorize transactions automatically or manually.
- Allow monthly and category-wise budget creation.
- Calculate total income, expenses, savings, and remaining budget.
- Generate alerts for unusual or high spending.
- Provide personalized recommendations using transaction patterns.
- Display financial information through simple charts and dashboards.
- Protect user information through authentication and access control.

2.3 Advantages



Feature	Benefit

Expense tracking	Improves awareness of daily spending
Budget planning	Helps users set spending limits
AI recommendations	Provides personalized suggestions
Dashboard	Makes financial patterns easier to understand
Alerts	Highlights budget risks early

3. EXISTING AND PROPOSED SYSTEM

3.1 Existing System

Traditional methods include notebooks, spreadsheets, calculator-based tracking, and simple expense applications. These methods can be useful for recording transactions but may require manual analysis. Some systems also provide generic tips that are not based on an individual user's spending history.

3.2 Proposed System

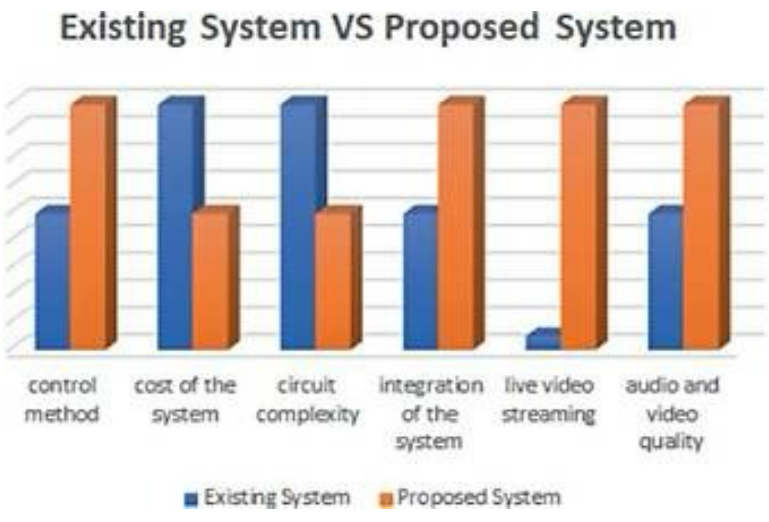
PocketSmart AI provides an integrated workflow. A user creates an account, enters financial transactions, sets budgets, and views a dashboard. The recommendation engine analyzes the available data and presents suggestions in simple language.

3.3 Comparison

Aspect	Traditional Tracking	PocketSmart AI
Transaction recording	Manual	Manual + optional assisted entry
Budget calculation	Manual	Automatic
Spending analysis	Manual	Automatic
Recommendations	Usually generic	Personalized
Visualization	Limited	Dashboard charts
Alerts	Manual checking	Rule-based alerts

3.4 Main Users

- Individual users
- Students
- Employees
- Families managing a personal monthly budget



4. REQUIREMENTS & FEASIBILITY STUDY

4.1 Functional Requirements

- User registration and login.
- Add, edit, delete, and view transactions.
- Create and update budget categories.
- Calculate monthly summaries.
- Generate charts and reports.
- Generate recommendations and budget alerts.
- Search and filter transactions.
- Export reports where supported.

4.2 Non-Functional Requirements

- Simple and responsive user interface.
- Reasonable response time for normal operations.
- Secure authentication and database access.
- Data consistency and backup support.
- Maintainable and modular source code.



4.3 Suggested Technology Stack

Layer	Suggested Technology
Frontend	HTML, CSS, JavaScript / React
Backend	Python Flask or FastAPI
Database	MySQL / PostgreSQL / SQLite
AI / Analytics	Python, Pandas, scikit-learn
Charts	Chart.js / Matplotlib
Development	Visual Studio Code
Version Control	Git

4.4 Feasibility

The project is technically feasible because the required technologies are widely available. It is economically feasible for an academic prototype because open-source frameworks and databases can be used. It is operationally feasible because the workflow is based on familiar actions such as entering transactions and setting budgets.

5. SYSTEM ARCHITECTURE & MODULES

The system follows a layered architecture. The presentation layer handles the user interface. The application layer processes business logic. The AI and analytics layer calculates patterns and recommendations. The data layer stores users, transactions, categories, budgets, and recommendation records.

5.1 Architecture Flow

User → Web/Mobile Interface → Authentication & API → Budget/Transaction Services → Database

↓

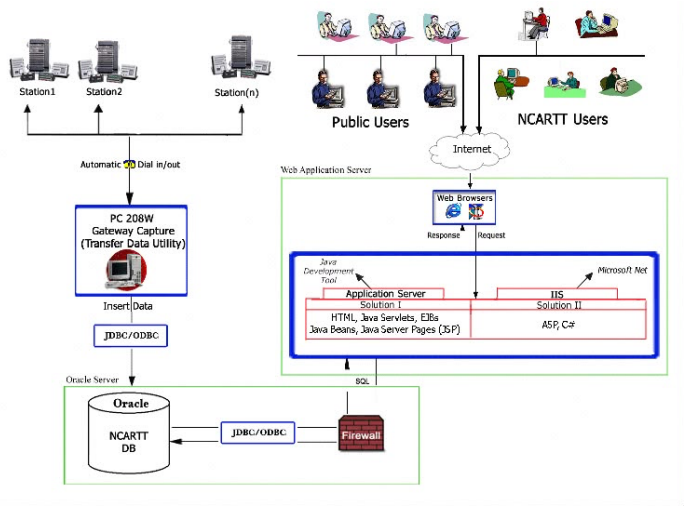
Analytics Engine

↓

Recommendation Engine

↓

Suggestions / Alerts



5.2 Major Modules

Module	Responsibilities
Authentication	Registration, login, session/token management
Transaction	Add and manage income/expense records
Budget	Create limits and calculate remaining amounts
Dashboard	Display summaries and charts
Analytics	Identify trends and spending patterns
Recommendation	Generate personalized suggestions
Notification	Show budget and spending alerts
Admin/Settings	Manage categories and application settings

5.3 Basic User Flow

Login → Dashboard → Add Income/Expense → Categorize Transaction → Set Budget → View Analysis → Receive Recommendation → Review Progress

6. DATABASE DESIGN

The database stores structured financial information. A relational database is suitable for the academic prototype because users, transactions, categories, budgets, and recommendations have clear relationships.

6.1 Main Tables

Table	Important Fields
users	user_id, name, email, password_hash, created_at
categories	category_id, user_id, category_name, type
transactions	transaction_id, user_id, category_id, amount, type, date, note
budgets	budget_id, user_id, category_id, amount, month, year
recommendations	recommendation_id, user_id, message, category, priority, created_at

6.2 Relationships

- One user can have many transactions.
- One user can create many budgets.
- One category can contain many transactions.
- A budget can be linked to a category and a user.
- One user can receive many recommendations.

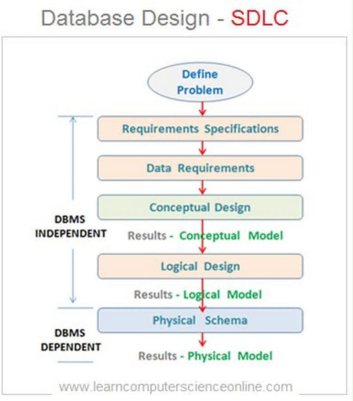
6.3 Example Calculation

Monthly Savings = Total Income − Total Expenses

Budget Remaining = Category Budget − Category Spending

Budget Usage (%) = (Category Spending / Category Budget) × 100

These calculations are used by the dashboard and recommendation engine.



7. AI RECOMMENDATION METHODOLOGY

The recommendation component is the main intelligent feature of PocketSmart AI. For an academic prototype, a hybrid approach can be used: rule-based recommendations provide predictable results, while machine-learning or statistical methods can be added when enough historical data is available.

7.1 Rule-Based Recommendations

- If category spending exceeds 80% of the budget, show a budget warning.
- If spending exceeds the category budget, suggest reviewing that category.
- If repeated transactions occur for a subscription category, show a recurring-expense summary.
- If monthly savings decrease compared with previous months, show a savings-awareness message.
- If an unusual transaction is detected by a chosen threshold or statistical method, request user review.

7.2 Optional Machine Learning

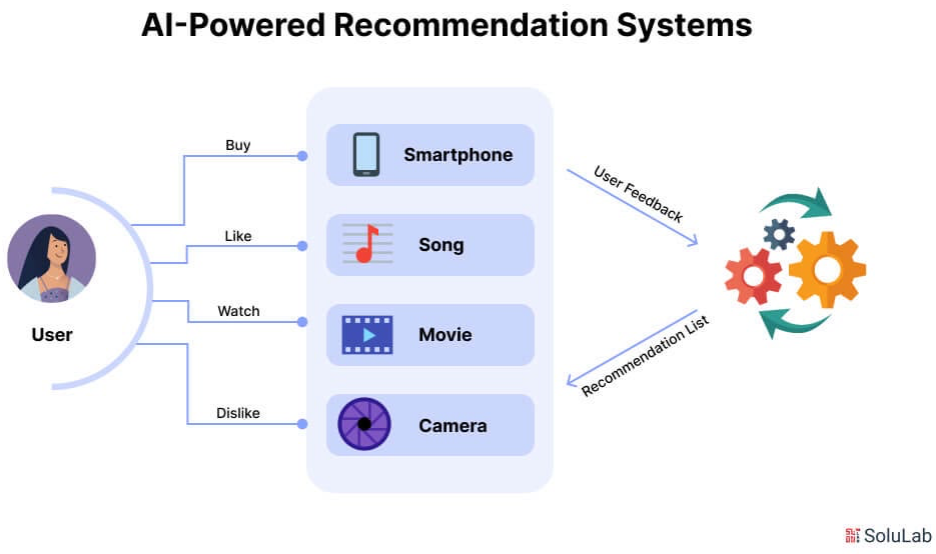
Historical transaction data can be transformed into features such as category totals, transaction frequency, average amount, monthly spending, and percentage of income. A clustering model can group spending patterns, while anomaly-detection techniques can flag transactions that differ significantly from normal behavior.

7.3 Recommendation Example

Input: Food budget = ₹8,000; current food spending = ₹7,200; monthly period is not yet complete.

System output: “Your food category has used about 90% of its planned budget. Review upcoming food expenses before the month ends.”

The recommendation is informational and should not be treated as professional financial advice.



8. USER INTERFACE & WORKFLOW

8.1 Login Page

The login page accepts the registered email/username and password. Authentication should be implemented securely using password hashing and protected sessions or tokens.

8.2 Dashboard

The dashboard presents total income, total expenses, estimated savings, budget usage, recent transactions, and recommendation cards. Pie charts or bar charts can be used to display category-wise spending.

8.3 Add Transaction Page

The user selects transaction type, enters amount, chooses a category, selects the date, and optionally adds a note. Validation prevents invalid or missing values.

8.4 Budget Page

Users can create a monthly limit for categories such as Food, Transport, Education, Shopping, Bills, and Entertainment.

8.5 Recommendation Page

Recommendations are displayed using simple messages with a category and priority. Users can review the reason for a recommendation and take action based on their own preferences.

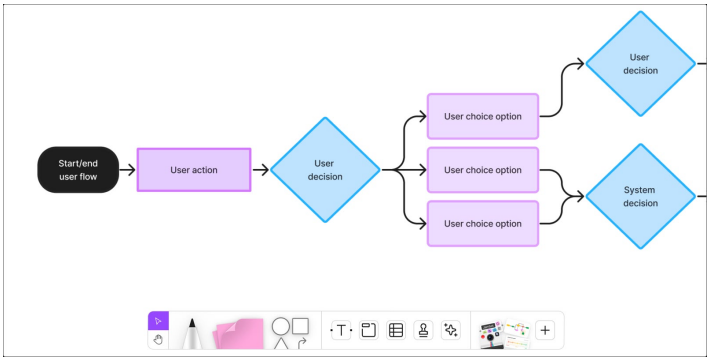
8.6 Sample Dashboard Layout

[Total Income] [Total Expenses] [Savings]

[Budget Usage Chart] [Category Spending Chart]

[Recent Transactions]

[AI Recommendations]



9. TESTING, SECURITY & PRIVACY

9.1 Testing Strategy

Test Case	Expected Result
Valid login	User reaches dashboard
Invalid login	Error message is shown
Add valid expense	Transaction is stored
Negative/invalid amount	Validation prevents submission
Create budget	Budget appears in dashboard
Budget reaches warning threshold	Recommendation/alert is generated
Delete transaction	Selected record is removed after confirmation
Unauthorized data access	Request is rejected

9.2 Security Measures

- Store passwords using secure hashing; never store plain-text passwords.
- Validate and sanitize user input.
- Use parameterized database queries or an ORM.
- Protect API endpoints with authentication and authorization.
- Use HTTPS in production.
- Avoid exposing sensitive information in error messages.

9.3 Privacy

Financial information is sensitive. The application should collect only necessary data, clearly explain how data is used, provide appropriate account controls, and avoid sharing personal transaction information without permission. The prototype should use sample data during demonstrations whenever possible.



10. RESULTS, CONCLUSION & FUTURE SCOPE

10.1 Expected Results

- Users can maintain organized income and expense records.
- Users can create category-wise budgets.
- The dashboard provides a clear view of financial activity.
- The recommendation engine identifies budget-related patterns.
- Alerts can help users notice high spending earlier.

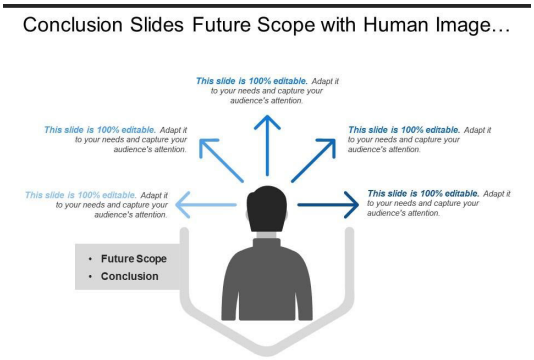
10.2 Conclusion

PocketSmart AI demonstrates how budgeting and intelligent recommendation techniques can be combined in a single personal finance application. The system reduces the effort required to calculate spending summaries and provides understandable recommendations based on user-entered financial information.

The project also demonstrates practical concepts in web application development, database management, data analysis, visualization, and artificial intelligence. With appropriate privacy and security controls, the system can be extended into a more complete personal budgeting assistant.

10.3 Future Scope

- Bank statement import with user permission.
- Receipt scanning and OCR for transaction entry.
- Natural-language chatbot for budget questions.
- Improved personalized models using longer-term spending history.
- Multi-language support, including Indian languages.
- Family/shared budgeting with controlled access.
- Goal-based saving plans.
- Mobile application version.
- More advanced anomaly detection and forecasting.



10.4 References / Learning Resources

- Python documentation – programming and data processing concepts.
- Flask/FastAPI documentation – web application and API development.
- MySQL/PostgreSQL documentation – relational database concepts.
- Pandas and scikit-learn documentation – data analysis and machine learning.
- Chart.js documentation – web-based data visualization.
-